

≡ Hide menu

- Introduction to Multilinear Regression:
Polynomial Regression Type
- Bias-Variance Trade-Off
- Multiple Linear Regression Model Optimization
- Slide Decks
- Assessments
- 🟢 Quiz: Week 2 Quiz

Submitted
- 🟢 Programming Assignment: Week 2: Multiple Linear Regression

3h
- 🔒 Peer-graded Assignment: Week 2: Multiple Linear Regression

Grading in progress
- 🔒 Review Your Peers: Week 2: Multiple Linear Regression

Peer-graded Assignment: Week 2: Multiple Linear Regression

Reviews 4 left to complete

Your fellow learner has submitted their assignment anonymously and your review will be anonymous to them. All names are still visible to course instructors.

Module 2 Peer Review

 by Anonymous Learner
November 12, 2023

👍 Like F Flag this submission

PROMPT

In **question 1d** in your Module 2 Jupyter notebook lab, you trained models with polynomial degrees up to 20, printing out each polynomial degree and r-squared value. Inspect the model summary from each model and discuss your findings. You filled in the highest order model that makes sense in sound_degree. Why don't higher-order models make sense?

Type your answer in the rich text field.

The higher order models (models greater than power = 2) did not make sense as the predictors had p-values that were not statistically significant.

While these models had higher r squared values, none of the predictors were statistically significant indicating the predictors were not required in the model.

RUBRIC

The **correct answer** for "Why don't higher-order models make sense?" is:

When you inspect the model.summary() for each model, models with a polynomial degree of 3 or higher have large p-values.

☒ 5 pts

Answer is based on the model summaries **and** states that higher polynomial degrees have large p-values.

☐ 2 pts

Answer is based on model summaries **but** does not correctly state that higher polynomial degrees have large p-values.

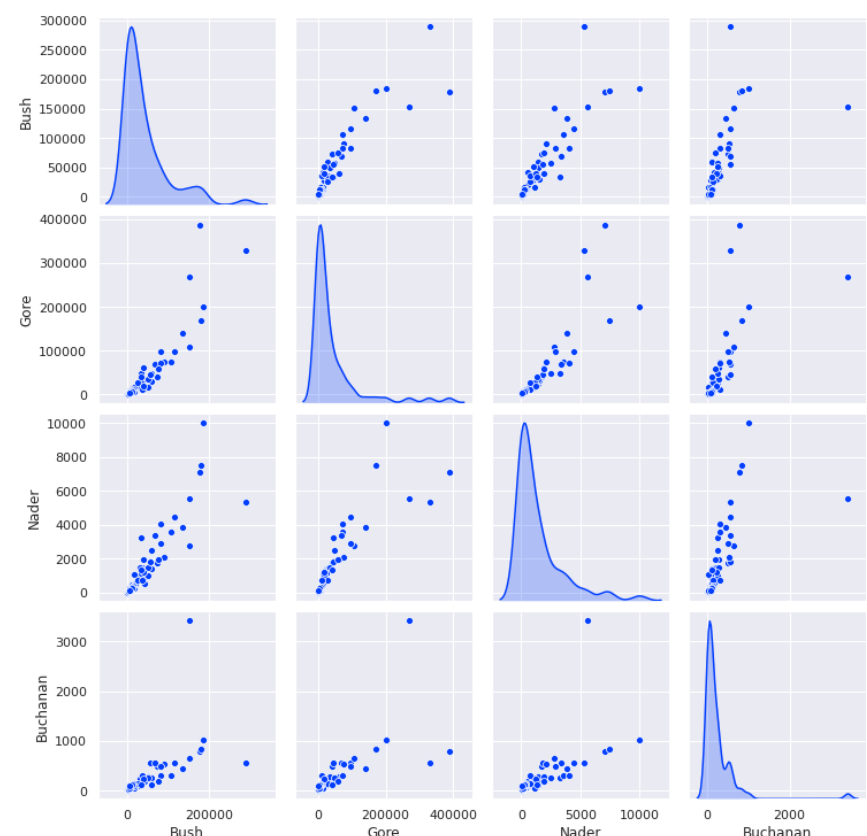
☐ 0 pts

No answer given, or gives answer that is not based on the model summaries.

PROMPT

In question 2a, you created a **pair plot** of the data using the seaborn library. Please upload a screenshot or copy of your pair plot.

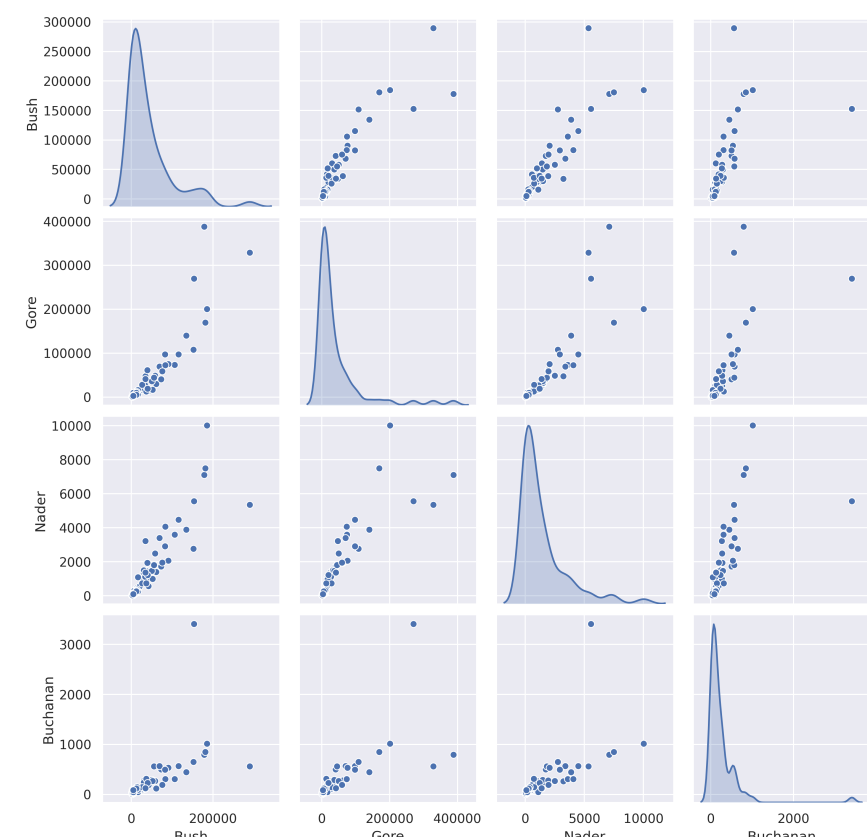
Pair Plot



Pair plot for module 2

RUBRIC

Here is a **sample pair plot solution**:



Did the learner submit a proper pair plot of the data?

☒ 3 pts

Author submitted a pair plot with voting data for Bush, Gore, Nader, and Buchanan.

☐ 1 pt

Author's pair plot is not similar to sample solution (e.g., doesn't include Bush, Gore, Nader, and Buchanan, or the vote totals)

☐ 0 pts

No pair plot attempt submitted.

PROMPT

After creating a pair plot of the data, question 2b has the following directions:

Comment on the relationship between the quantitative datasets. Are they **correlated**? Are they **collinear**?

Type your answer in the rich text field.

The dataset has columns that are highly correlated. For eg., the correlation between the Bush column and the other columns is as follows

1. Gore - 0.9126

2. Nader - 0.8922

3. Buchanan - 0.6251

Gore is correlated to Nader and Buchanan as follows

1. Nader - 0.8643

2. Buchanan - 0.6913

And finally, Nader and Buchanan have a correlation of 0.6542

Collinearity is present in the dataset as can be seen from the Variance Inflation Factors (VIF) for the model. The VIF values of concern are as follows

1. Gore - 4.441

2. Nader - 4.053

Buchanan has a VIF of 1.963 which while greater than the acceptable value of 1 is manageable.

In summary, the dataset is correlated and has columns that are collinear so models created from this dataset might not be very useful.

RUBRIC

Here is a sample solution:

They are correlated but not collinear. You can use the correlation coefficient to see that e.g. the Bush v. Gore data has the highest correlation coefficient at 0.91.

Evaluate the learner's solution. The learner's solution does not need to match the sample. Given the subjective nature of the question, award full points as long as the learner answers the question and adequately explains their reasoning.

☒ 2 pts

Includes both of the following: answers question and explains reasoning.

☐ 0 pts

Missing one or both of the following: question answer or explanation of reasoning.

PROMPT

Question 2c has the following directions:

Construct a multi-linear model called **model** without interaction terms. Predict the Bush column on the other columns and print out the summary table. Use the full data (not train-test split for now) and do not scale features.

Is there any insignificant feature(s)? Explain your answer.

Type your answer in the rich text field.

When Bush is predicted against other columns without interactions (ie formula = Bush ~ Gore + Nader + Buchanan), the **Buchanan** predictor has a p-value of **0.363** which is **NOT** statistically significant indicating this predictor can be removed from the model.

RUBRIC

The correct answer is: Buchanan has a large p-value therefore it is insignificant.

☒ 2 pts

Correctly identifies that Buchanan is the only insignificant feature due to having a large p-value.

☐ 1 pt

Answer partially correct and **includes at least one of the following:** identifies that Buchanan is insignificant **or** identifies that only one feature is insignificant **or** identifies that Buchanan is insignificant due to having a large p-value.

☐ 0 pts

Answer not submitted or fully incorrect: Does not identify that Buchanan is insignificant **and** incorrectly identifies other features as being insignificant **and** does not include that Buchanan is insignificant due to having a large p-value.

PROMPT

Question 2d has the following directions:

Construct a multi-linear model with interactions that are statistically significant at the p = 0.05 level. You can start with full interactions and then eliminate interactions are do not meet the p = 0.05 threshold. Name this model object **model_multi**. Upload a screenshot of your statsmodels OLS results summary so your peers can see which model you constructed.

Multiple Linear Regression with Interaction

```
ols > summary()
OLS Regression Results
Dep. Variable:      bush    R-squared:      0.954
Model:              OLS    Adj. R-squared:  0.944
Method:             Least Squares    F-statistic:  222.3
Date:               Sat, 11 Nov 2023    Prob (F-statistic): 1.48e-42
Time:               20:24:17          [OmitTED]
No. Observations:    47
Df Residuals:        45
Df Model:             2
Covariance Type:     nonrobust

               coef    std err          t      P>|t| [0.025   0.975]
-----
Intercept    13.3893    3893.361     0.034    0.969   -7467.457    15198.229
Gore         -0.8708    0.095    -9.272    0.000    -1.060   -0.682
Nader        -0.7608    0.092    -8.368    0.000    -0.943   -0.578
Gore*Nader   -7.305e-05  1.45e-05   -5.090    0.000    -0.000   -4.76e-05
R-squared:      0.954
Adj. R-squared:  0.944
F-statistic:    222.3
Prob (F-statistic): 1.48e-42
AIC:            11.480
BIC:            11.514
Kurtosis:       10.979
Cond. No.:      4.74e+06
```

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.74e+06. This might indicate that there are strong multicollinearity or other numerical problems.

RUBRIC

Here is a sample solution (Bush~Gore+Nader*Buchanan). :

OLS Regression Results

Dep. Variable:	Bush	R-squared:	0.910			
Model:	OLS	Adj. R-squared:	0.904			
Method:	Least Squares	F-statistic:	155.9			
Date:	Mon, 01 Nov 2021	Prob (F-statistic):	1.30e-31			
Time:	22:33:35	Log-Likelihood:	-747.89			
No. Observations:	47	AIC:	1506.			
Df Residuals:	42	BIC:	1517.			
Df Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	-1980.0202	3516.613	-0.563	0.575	-9010.132	5049.088
Gore	0.4302	0.061	7.024	0.000	0.307	0.553
Nader	16.2123	2.350	6.898	0.000	11.514	20.910
Buchanan	64.4268	16.584	3.885	0.000	31.278	97.580
Nader*Buchanan	-4.0141	0.033	-4.735	0.000	-4.020	-4.008
Omnibus:	21.432	Durbin-Watson:	1.966			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	177.885			
Skew:	-0.162	Prob(SB):	0.26436			
Kurtosis:	10.979	Cond. No.	4.74e+06			

Other possible solutions include:

Intercept+Gore+Gore*Nader+Nader

Intercept+Buchanan+Buchanan:Gore+Gore

Intercept+Buchanan+Buchanan:Nader+Gore+Nader

Did the learner submit a multi-linear model with interactions that are statistically significant at the p = 0.05 level?

☒ 10 pts

The author submitted the sample solution or one of the other possible solutions. This means that the learner submitted a multi-linear regression model with interactions statistically significant at the p = 0.05 level.

